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Criteria for Extending Preventive Maintenance Plans in Power Substations Using Reliability Block Diagrams

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Abstract

Maintaining aging equipment is essential for ensuring the reliability and performance of power substations, which are critical components of electric power systems. Preventive maintenance (PM) policies are widely used in this context, but their definition is often based on fixed schedules that do not explicitly account for system degradation and failure dynamics. This paper proposes a structured decision-making framework for evaluating the extension of PM intervals in complex engineering systems. The approach integrates Reliability Block Diagram (RBD) modeling with an imperfect maintenance representation based on the Generalized Renewal Process (GRP) and a Kijima model, allowing the representation of cumulative degradation effects. A decision criterion combining system reliability and the expected number of failures is defined to assess the feasibility of maintenance strategies. The framework is applied to a power substation case study, with emphasis on the transmission line subsystem, considering different PM extension scenarios and parameter uncertainty through sensitivity analysis. The results show that maintenance interval extension must be treated as a constrained problem, in which reductions in maintenance effort must be balanced against increased degradation and failure risk. For the presented case study, only moderate extensions are found to be consistently feasible across all evaluated conditions. The proposed approach provides a systematic and practical method for supporting maintenance planning under uncertainty, enabling consistent and transparent evaluation of maintenance strategies.

Keywords: reliability block diagram; RBD; preventive maintenance; maintenance interval extension; imperfect maintenance; decision-making framework

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1. Introduction

The ongoing transformation of the electrical power sector, driven by new generation technologies and evolving market demands, has increased the importance of efficient and reliable operation of transmission and distribution systems. In this context, maintenance planning remains a critical activity for ensuring system performance while meeting economic and operational constraints [1]. Utilities must continuously balance reliability targets, resource allocation, and operational risks to maintain system functionality at an acceptable level.

Power substations play a central role in this process, as they comprise multiple interconnected components with distinct degradation mechanisms and operational characteristics. Failures in these systems may lead to extended outages, high repair costs, and

significant impacts on service continuity. Therefore, maintenance planning must rely on structured approaches capable of identifying critical components and evaluating the impact of maintenance actions on system performance [1–3].

Reliability analysis is a fundamental tool in this context. Among the available methods, the Reliability Block Diagram (RBD) is widely used to represent system structure and functional dependencies, enabling the evaluation of system reliability and availability based on component behavior [4–6]. RBD models also support the identification of critical components and the assessment of system performance under different operating conditions.

In practice, however, maintenance planning in power substations is still largely based on predefined preventive maintenance (PM) schedules, typically implemented as time-based policies in which intervention intervals are defined according to operational procedures and engineering practice. Although condition-based approaches have been increasingly introduced, PM remains the predominant strategy in transmission systems [7].

Maintenance intervals are commonly derived from a combination of manufacturer recommendations, operational experience, and utility-specific guidelines [8], which may not fully reflect the actual degradation behavior of individual components or the specific operating conditions of a given substation [9]. As a result, the relationship between maintenance intervals and system degradation dynamics is not explicitly modeled, and the impact of maintenance actions on system reliability is not always quantitatively assessed.

This raises a central engineering question: to what extent can PM intervals be extended without compromising system reliability or increasing the risk of failure?

Addressing this question requires approaches that explicitly link system reliability modeling, maintenance representation, and decision-making criteria. In this work, a structured framework is proposed that combines RBD-based reliability modeling with an imperfect repair formulation based on the Generalized Renewal Process (GRP) and the Kijima Type II model [10–12], enabling the representation of cumulative degradation effects. In addition, a decision-making criterion based on system reliability and the expected number of failures is defined to support the evaluation of maintenance strategies.

The main contribution of this work lies in the integration of RBD-based reliability modeling, GRP-based imperfect maintenance modeling, and a structured decision-making criterion combining system reliability and the expected number of failures within a unified framework. This integration enables the systematic evaluation of maintenance interval extension scenarios while explicitly accounting for degradation effects, maintenance actions, and decision constraints.

1.1. Research Gap and Motivation

Despite its widespread use in reliability assessment, the application of RBD models as decision support tools for maintenance planning remains limited in scope. Traditional reliability modeling approaches are typically formulated under static assumptions, considering constant operating conditions and time-invariant system behavior. However, real-world engineering systems are subject to dynamic influences such as varying operational conditions, environmental factors, and maintenance actions, which significantly affect system reliability [13].

In many practical implementations, RBD-based analyses are restricted to static evaluations of reliability or availability, without explicitly incorporating the effects of maintenance actions on system degradation over time. In addition, classical RBD formulations are not designed to represent dynamic behaviors, dependencies, or interactions among system components, limiting their applicability in complex and evolving systems [14].

Moreover, these models often rely on aggregated historical data and do not account for the actual condition of assets, as obtained from monitoring, diagnostic, or prognostic infor-

mation [15]. As a result, current reliability modeling approaches are inherently limited in their ability to support maintenance decision-making in dynamic operating environments.

Another limitation concerns the weak connection between reliability modeling outputs and maintenance decision-making. Although reliability models provide quantitative indicators such as reliability and availability, these metrics are not always directly linked to structured criteria for evaluating maintenance strategies [15]. In practice, maintenance planning often depends on predefined schedules or heuristic criteria, rather than on a systematic assessment of the impact of maintenance decisions on system performance.

These limitations reveal a gap between reliability modeling and practical maintenance decision-making. In particular, there is a need for approaches that explicitly integrate (i) system reliability modeling, (ii) realistic representations of maintenance actions under imperfect repair assumptions, and (iii) decision criteria capable of supporting the evaluation and comparison of maintenance strategies. This work addresses this gap by proposing a unified framework that combines RBD-based reliability modeling, GRP-based imperfect maintenance, and a constraint-based decision-making criterion.

1.2. Literature Review

Reliability assessment of electrical power systems has been extensively studied, particularly in the context of transmission and distribution substations. In these systems, reliability is associated with the ability to continuously supply electrical energy within acceptable quality standards, being directly influenced by equipment performance, maintenance practices, and system configuration [16–18]. Substation reliability studies typically focus on equipment failures, preventive and corrective maintenance (CM) actions, and their impact on power interruptions.

Power substations represent critical assets in electrical networks, and their performance is strongly dependent on appropriate maintenance planning. Failures in these systems can result in significant operational and economic impacts, motivating the use of reliability modeling and data analysis techniques to support maintenance strategies [1]. In this context, PM plays a central role, but determining how maintenance intervals can be adjusted without compromising system reliability remains a relevant and open question.

RBD are widely used to represent system structure and evaluate reliability and availability in complex engineering systems. Their graphical formulation enables the identification of critical components and supports the analysis of system behavior based on component failure and repair characteristics [19]. Applications of RBD in power systems include the evaluation of redundancy strategies, failure probabilities, and asset management decisions, demonstrating their usefulness for system-level analysis [5,20–22].

However, traditional reliability modeling approaches are often based on simplified assumptions, such as constant failure and repair rates, component independence, and static operating conditions [5]. These assumptions limit the ability of such models to represent real system behavior, particularly in the presence of degradation processes, varying environmental conditions, and maintenance actions over time.

In parallel, maintenance strategies in power systems have evolved from corrective and time-based preventive approaches toward more advanced techniques, including condition-based and predictive maintenance. These approaches rely on monitoring technologies, data acquisition, and analytical methods to improve maintenance planning and system performance. For instance, thermographic inspections and data-driven techniques have been successfully applied to detect degradation and support maintenance decisions in substations [23,24].

Recent developments in power system inspection have been driven by the integration of advanced inspection platforms, including unmanned aerial systems and robotic solutions,

which enable automated data acquisition and reduce dependence on manual inspections. Robotic inspection systems with integrated navigation and defect management capabilities have been proposed to improve inspection efficiency [25], while UAV-based approaches expand monitoring capabilities in large-scale and hard-to-access environments [26].

At the methodological level, artificial intelligence and data-driven techniques have significantly enhanced fault detection and diagnosis capabilities. These approaches structure inspection processes into stages such as data acquisition, component detection, and fault diagnosis, achieving improved performance compared to traditional methods [27].

Additionally, reliability-centered maintenance approaches have been proposed to integrate reliability analysis, maintenance costs, and risk into structured maintenance strategies, improving system performance when compared to traditional time-based approaches [7]. These advances are closely associated with the transition toward predictive maintenance strategies, in which monitoring, diagnostic, and prognostic data are increasingly being integrated into reliability models to support maintenance decision-making [15].

Despite these advances, maintenance planning in practical applications is still largely based on predefined schedules and heuristic criteria, as traditionally established in maintenance practice [28,29]. More recent studies indicate that this paradigm remains prevalent, as current reliability models still present limited integration with real-time asset condition and decision-making processes [15].

While reliability models provide quantitative indicators such as reliability and availability, their use in supporting maintenance decision-making remains limited, particularly due to the lack of integration with asset condition and operational data [15]. In addition, traditional reliability modeling approaches are often based on simplified assumptions and do not explicitly capture the cumulative effects of maintenance actions on system degradation or the dynamic behavior of components [14]. As a result, these models provide limited support for the evaluation of maintenance strategies in dynamic operating environments [13].

As a result, although reliability modeling and maintenance optimization have been widely studied, there is still a lack of integrated approaches that combine system-level reliability analysis, realistic maintenance modeling, and structured decision-making criteria. This limitation reinforces the need for frameworks capable of systematically evaluating maintenance interval extension scenarios in complex engineering systems.

The remainder of this paper is organized as follows. Section 2 presents the proposed methodology, including the reliability modeling approach, the representation of maintenance actions based on imperfect repair, and the decision-making criteria adopted. Section 3 describes the application example, including the system under study, input data, and simulation setup. Section 4 discusses the results, focusing on the evaluation of maintenance interval extension scenarios and their impact on system reliability and expected number of failures. Finally, Section 5 presents the main conclusions and outlines directions for future work.

2. Methodology

The proposed framework follows a structured workflow comprising system modeling, maintenance representation, simulation of maintenance scenarios, and evaluation based on predefined decision criteria for assessing the feasibility of PM interval extensions. The approach integrates RBD modeling, probabilistic failure modeling, and repairable system theory to represent system behavior under different maintenance strategies.

The overall methodological framework is illustrated in Figure 1, which summarizes the main steps of the approach, including system modeling, parameter definition, simulation of maintenance scenarios, and decision-making based on reliability and failure criteria.

This structure enables the consistent evaluation of the impact of maintenance interval modifications on system performance.

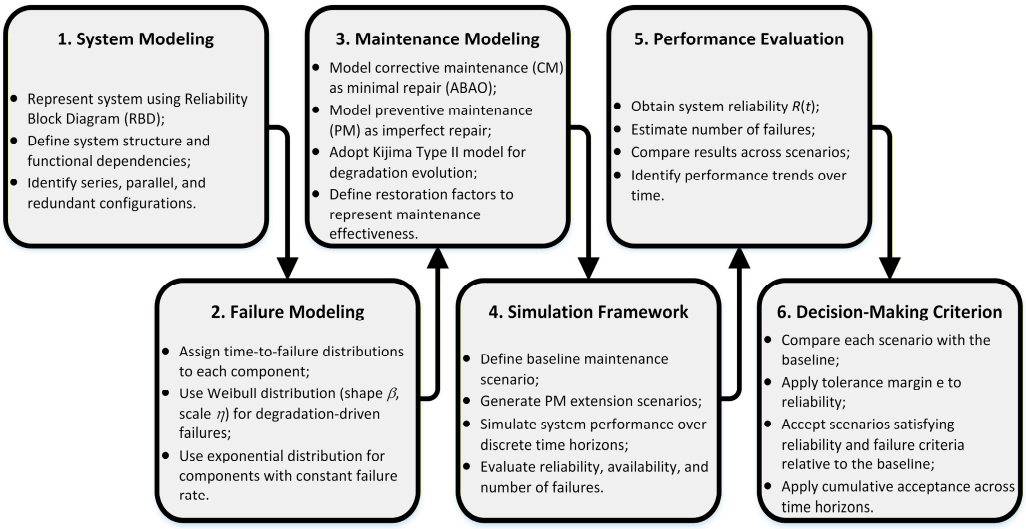


Figure 1. Methodological framework for reliability-based evaluation of PM interval extensions.

The methodology supports the comparison of alternative maintenance strategies with respect to a baseline configuration, allowing the identification of feasible extensions that do not compromise system reliability or increase failure occurrence beyond acceptable limits.

2.1. Reliability Block Diagram Fundamentals

RBD are widely used to represent the functional structure of engineering systems and to evaluate system reliability based on the configuration and interaction of their components. In this framework, each block represents a component that can be in either an operational or failed state, and system reliability is determined by the logical relationships among these components.

Component reliability is described through time-to-failure probability distributions, such as the exponential distribution [22] or the Weibull distribution [6], which capture different failure behaviors. System reliability is then obtained by combining the individual component reliabilities according to the system configuration.

Typical RBD configurations include series, parallel, k-out-of-N, and standby redundancy structures, as illustrated in Figure 2. These configurations define how component failures propagate to the system level and allow the representation of different redundancy strategies [30].

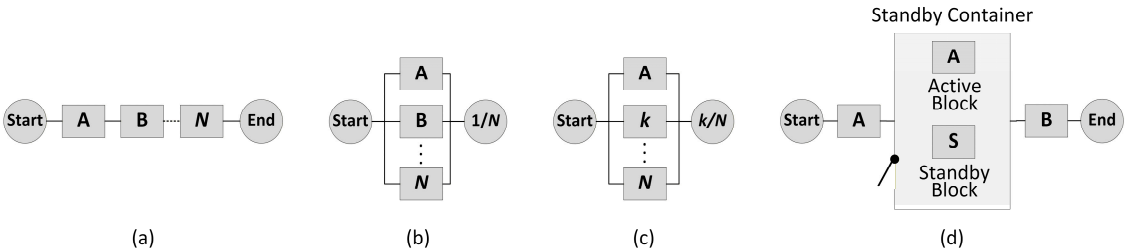


Figure 2. RBD configuration types: (**a**) Series system; (**b**) parallel (redundant) system; (**c**) *k*-out-of-*N* system; and (**d**) standby redundancy system with switching mechanism between active and standby units.

In addition to reliability, RBD models can incorporate maintainability aspects, enabling the evaluation of system availability by considering repair processes and downtime [31].

In the present work, the RBD is used as the structural backbone for integrating component-level failure behavior and maintenance effects within a system-level reliability framework. This representation enables the consistent evaluation of system performance under different maintenance strategies.

2.2. Failure and Reliability Modeling

The reliability behavior of the components represented in the RBD is modeled through time-to-failure probability distributions. In reliability analysis, the selected distribution must be capable of representing the degradation characteristics associated with the failure mode of the component under analysis. Among the available models, the Weibull distribution is widely used due to its flexibility in describing different failure behaviors and is adopted in this work as the main reliability model [6]. Its reliability function is given by Equation (1).

$$R(t) = e^{-\left(\frac{t}{\eta}\right)^{\beta}} \quad (1)$$

where $R(t)$ is the reliability, η is the characteristic life, and β is the shape parameter. When $\beta < 1$, the failure rate decreases with time, which is typically associated with early-life failures. When $\beta = 1$, the failure process becomes memoryless, and the failure rate is constant. When $\beta > 1$, the failure rate increases with time, which is consistent with degradation-driven and wear-out failure modes.

The Weibull distribution is particularly suitable for power substation systems, as different components may exhibit distinct degradation patterns over their operational life. This is especially relevant in the present study, where the objective is to evaluate the impact of PM interval extensions on system reliability. In this context, the Weibull model enables the representation of time-dependent failure behavior and supports the analysis of interactions between degradation processes and maintenance policies. This capability is essential for coupling failure behavior with the GRP-based maintenance modeling adopted in this work.

For components whose failure behavior is better represented by a constant failure rate, such as some electronic systems, the exponential distribution is adopted [22]. This model is appropriate when no significant aging effect is assumed, meaning that the conditional probability of failure remains constant over time.

In addition to reliability modeling, maintainability must also be represented in the RBD simulation. Repair durations are modeled using a Normal distribution with component-specific mean values and a standard deviation of σ = 1 h. This assumption reflects the relatively stable nature of maintenance operations, in which repair times tend to vary around a characteristic value due to factors such as technician experience, spare part availability, environmental conditions, and site accessibility.

To ensure physical consistency, the distribution is truncated at zero, since repair times cannot be negative. The Normal distribution is adopted as an approximation for repair time variability, assuming symmetric dispersion around the mean, which is consistent with relatively standardized maintenance procedures.

This combination of distributions provides a consistent probabilistic basis for the RBD model: Weibull distributions are used to represent degradation-dependent failure behavior, exponential distributions are used for components with approximately constant failure rates, and Normal distributions are used to represent repair duration variability. Together, these models enable the integrated simulation of reliability, maintainability, and maintenance effects in the subsequent steps of the methodology.

2.3. Maintenance Modeling

Maintenance actions are modeled within the framework of repairable systems, in which components undergo successive failures and maintenance interventions throughout their operational life. In this context, maintenance does not necessarily restore the system to an "as good as new" (AGAN) condition, and different types of interventions must be distinguished according to their effect on system degradation.

CM is modeled as a minimal repair process, in which the system is restored to an operational condition without modifying its accumulated degradation state. This corresponds to the classical "as bad as old" (ABAO) assumption, where the system, immediately after repair, resumes operation in a condition equivalent to that just before failure [11].

PM, on the other hand, is modeled as an imperfect repair process, in which maintenance actions partially reduce the accumulated degradation of the system. In real systems, maintenance interventions rarely achieve full restoration, and the post-maintenance condition typically lies between AGAN and ABAO, depending on the effectiveness of the intervention [32].

To represent this behavior, the present work adopts a formulation based on the concept of virtual age within the framework of the Generalized Renewal Process (GRP) [12]. In this approach, the effect of maintenance is modeled through modifications to the system's effective age, which captures the cumulative impact of degradation and repair over time.

Among the available formulations, the Kijima Type II model is adopted [11,12]. In this model, maintenance actions affect the total accumulated degradation of the system, rather than only the degradation since the last intervention. As a result, the virtual age is updated as a function of the overall degradation history, allowing maintenance actions to proportionally reduce, but not eliminate, the system's effective age. This formulation reflects the cumulative nature of aging processes in substation equipment.

From a physical standpoint, this formulation is consistent with systems in which degradation accumulates over time and is only partially mitigated by maintenance actions [33]. This behavior is typical of electrical substation components, where aging mechanisms such as insulation degradation, mechanical wear, and environmental exposure are not fully reversible.

In practical implementation, the effectiveness of PM is represented through restoration factors, which quantify the extent to which maintenance actions reduce accumulated degradation. These factors are defined for each component based on maintenance procedures and engineering judgment and are used to parameterize the model within the simulation environment. To ensure robustness, their influence is assessed through sensitivity analysis.

This modeling approach explicitly separates functional restoration from degradation evolution. CM restores system operability without altering the accumulated degradation state, whereas PM reduces degradation through virtual age adjustment, delaying future failures. The combined representation results in a hybrid repairable system model that integrates minimal repair (CM) and imperfect repair (PM) within a unified GRP-based framework.

The modeling of failure and maintenance processes at the component level is illustrated in Figure 3, which represents the interaction between degradation, corrective actions, and preventive interventions within the RBD framework.

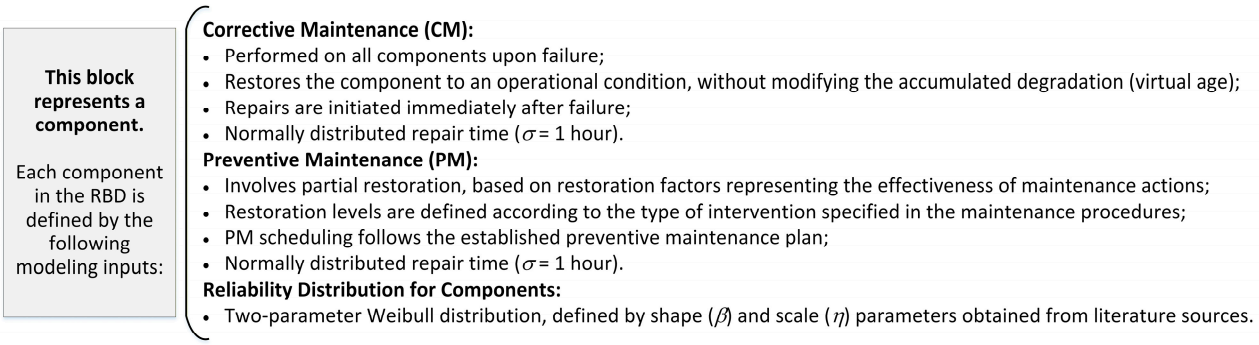


Figure 3. Component-level inputs for reliability and maintenance modeling in the RBD.

2.4. RBD Construction and Simulation

The construction of the RBD follows the definition of the system's functional structure, in which components are arranged according to their operational dependencies and logical relationships. Each block represents a component characterized by its reliability behavior and maintenance parameters, as defined in the previous subsections. The resulting RBD provides a structured representation of the system, enabling the evaluation of its performance under different operating and maintenance conditions.

The modeling process integrates three fundamental elements at the component level: (i) failure behavior, represented by time-to-failure distributions; (ii) CM actions, modeled as minimal repairs; and (iii) PM actions, modeled as imperfect repairs through restoration factors. This integration allows the representation of both degradation processes and maintenance effects within a unified framework.

Once the RBD is defined, system performance is evaluated through stochastic simulation. The analysis considers increasing simulation horizons, allowing the assessment of system behavior over progressively longer operating periods. For each scenario, key performance indicators are obtained, including system reliability $R(t)$, availability $A(t)$, and the expected number of failures. Results are estimated from repeated stochastic runs, ensuring statistical stability of the evaluated metrics.

The simulation framework is structured to enable the comparison of different maintenance strategies. A baseline scenario is first defined, representing the current PM policy. Alternative scenarios are then generated by modifying maintenance intervals, allowing the evaluation of the impact of maintenance extensions on system performance.

By maintaining the system structure and failure characteristics unchanged across scenarios, the simulation isolates the effect of maintenance policy variations. This approach ensures that observed differences in reliability and failure occurrence are directly attributable to changes in maintenance intervals.

This framework provides a consistent basis for evaluating the trade-offs associated with PM extensions, supporting the systematic comparison of candidate strategies under different operating horizons. The resulting performance metrics are subsequently used in the decision-making process described in the following subsection.

2.5. Decision-Making Criterion

The maintenance decision process is based on the evaluation of system reliability, $R(t)$, and the expected number of failures under different PM interval scenarios. Availability, $A(t)$, was not adopted as a decision variable.

Reliability represents the probability of failure-free operation over time and is directly associated with component degradation. In contrast, availability incorporates both failure behavior and maintenance-related downtime, including planned interventions. As a result, availability is inherently influenced by the maintenance policy itself. When PM intervals

are extended, the reduction in scheduled downtime mechanically increases availability, regardless of the underlying degradation process. Therefore, availability is not suitable for comparing maintenance strategies with different intervention frequencies.

The decision-making process is formulated as a constraint-based evaluation of reliability and failure occurrence. While extending PM intervals reduces maintenance effort, it increases the risk of failure. To ensure a conservative and risk-aware assessment, both criteria are considered simultaneously.

For each candidate scenario, system performance is evaluated through simulation, and the corresponding reliability and number of observed failures are obtained. These results are then compared with those of a reference scenario defined by the baseline maintenance strategy.

To avoid excessive sensitivity to small variations in reliability, a tolerance margin, denoted by ε, is introduced. This parameter defines the maximum acceptable deviation in reliability relative to the baseline and can be interpreted as a risk tolerance margin. Lower values of ε lead to more conservative decisions, requiring maintenance strategies to closely match baseline reliability levels, whereas higher values allow greater flexibility at the expense of increased exposure to failure risk. Therefore, the selection of ε should be aligned with operational requirements, acceptable risk levels, and system criticality.

Thus, a maintenance strategy is considered acceptable for a given simulation horizon if the conditions defined in Equation (2) are simultaneously satisfied relative to the baseline scenario.

$$R_s(t) \geq R_{baseline}(t) - \varepsilon \quad \text{and} \quad N_s < N_{baseline} \tag{2}$$

where $R_s(t)$ is the reliability of the evaluated scenario, $R_{baseline}(t)$ is the reliability of the baseline strategy, N_s is the number of observed failures, and $N_{baseline}$ is the corresponding value for the baseline.

The simulation horizons considered in this study are cumulative, meaning that each time horizon reflects system behavior over the entire operating period up to that point. Consequently, the decision process is applied progressively: if a maintenance strategy fails to satisfy the acceptance criteria at a given horizon, it is considered unacceptable for all subsequent horizons.

This approach ensures that maintenance interval extensions are only accepted when they do not introduce additional failure risk and maintain reliability within an acceptable margin. Table 1 summarizes the criteria used in the screening of maintenance strategies.

Table 1. Decision-making criterion.

Criterion	Accept (✔)	Reject (✘)
Reliability	$R_s(t) \geq R_{baseline}(t) - \varepsilon$	$R_s(t) < R_{baseline}(t) - \varepsilon$
Number of Failures	$N_s \leq N_{baseline}$	$N_s > N_{baseline}$

3. Case Study

The methodology presented in Section 2 is applied to a transmission substation in Brazil that has recently undergone expansion and modernization, resulting in changes in system configuration and installed capacity, making it suitable for evaluating PM interval extensions.

Figure 4 presents a schematic representation of the main systems found in transmission substations. These systems are consistent with those present in the analyzed substation and define the functional structure adopted in the reliability modeling.

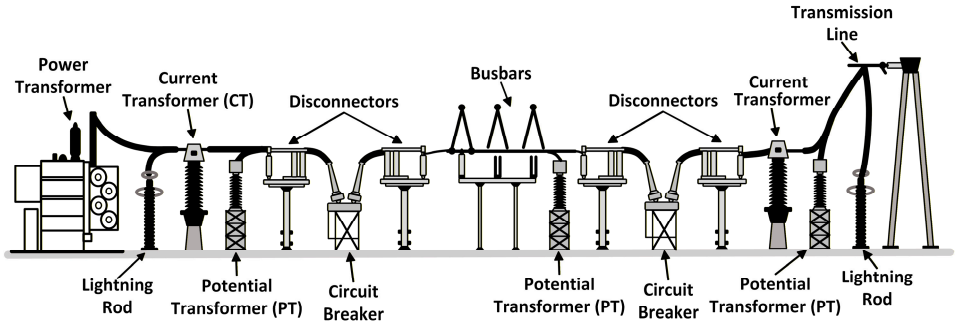


Figure 4. Substation main systems.

3.1. System Breakdown

The substation is decomposed into its main functional systems to support the construction of the RBD. Figure 5 presents the macro functional tree adopted in this work, and Table 2 summarizes the primary function of each system.

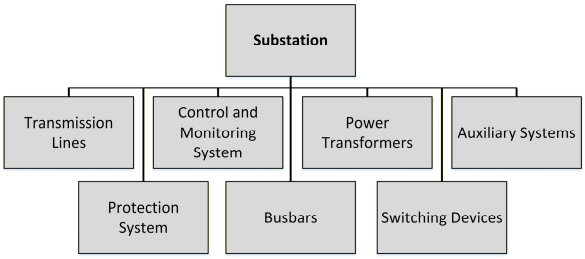


Figure 5. Substation functional tree.

Table 2. Substation functional description.

System	Primary Function
Substation	Receive, transform, and distribute electrical energy between transmission networks and load centers.
Transmission Lines	Receive electrical energy from generation units or transmission substations and deliver it to distribution networks or load centers.
Protection System	Detect abnormal conditions and initiate protective actions to ensure system safety and reliability.
Control and Monitoring System	Acquire, monitor, and control operational variables for system supervision and operation.
Busbars	Interconnect substation equipment, enabling power transfer and operational flexibility.
Power Transformers	Transform voltage levels within the power system.
Switching Devices	Perform switching operations, including interruption and isolation, under normal and fault conditions.
Auxiliary Systems	Supply AC and DC power to support protection, control, communication, and auxiliary services.

This functional decomposition is used to represent the substation at a system level, allowing the identification of the main elements that contribute to power transmission, transformation, control, and protection. These systems define the structural basis for the reliability modeling and the subsequent evaluation of maintenance strategies.

3.2. System Knowledge Acquisition

The development of the RBD requires a detailed understanding of the substation architecture, its operational objectives, and the functional dependencies among its main systems.

In this study, system knowledge was obtained through the analysis of technical documentation, including single-line diagrams, equipment specifications, and protection schemes. This information was complemented by maintenance records and operational data, which provided insights into equipment behavior, failure modes, and maintenance practices.

Additional understanding of system configuration and operational constraints was obtained through consultation with field experts and system operators. These sources were used to identify critical components and to ensure that the functional relationships represented in the RBD are consistent with the actual operation of the substation.

This knowledge acquisition process supports the construction of a representative system model, ensuring that the RBD structure and parameterization reflect the actual operational logic, constraints, and dependencies required for reliability evaluation and maintenance decision analysis.

3.3. Data Sources and Processing

Reliability and maintenance data were obtained from three complementary sources, combining structured maintenance procedures, operational records, and literature-based information.

First, maintenance procedures were collected from formal technical instructions adopted by the utility. These documents define PM activities, execution intervals, and standard intervention practices for each component, and were used to establish the PM frequency and maintenance policies applied in the model.

Second, operational data were obtained from substation work orders and event records, including maintenance logs, inspection reports, and disturbance records. These datasets provide information on event occurrence, classification (e.g., forced and scheduled outages), duration, and reported causes. They were used to characterize maintenance execution and intervention durations, ensuring consistency between modeled and observed system behavior.

However, the available operational data do not provide sufficient failure information for all critical components. This limitation is common in operation and maintenance (O&M) contexts, where historical records may be incomplete or insufficient for reliable statistical estimation of failure distributions. In addition, intensive PM policies tend to reduce the occurrence of observable failures, further restricting data availability. To address this limitation, literature-based parameters are adopted, providing a consistent and conservative representation of component reliability.

Therefore, reliability parameters were obtained from literature sources, including published studies and reliability databases for similar electrical equipment [34–36]. Representative Weibull parameters were adopted to provide a consistent and conservative description of component failure behavior, in accordance with established reliability modeling practices. These parameters are considered representative of typical operating conditions for high-voltage substation equipment.

The combination of these data sources enables a consistent parameterization of the RBD model. Maintenance procedures define PM policies, operational records support the representation of maintenance actions and their durations, and literature data provide the basis for failure modeling. This integrated approach ensures that the model reflects both the degradation behavior of the components and the practical operational conditions of the system.

3.4. Reliability Parameter Definition

The reliability parameters used in the RBD model were defined based on literature sources, as described in Section 3.3. Due to the limited availability of failure data for critical components, representative Weibull parameters were adopted from published studies and reliability databases for similar electrical equipment [34–36].

The Weibull distribution was selected due to its flexibility in representing different failure behaviors observed in power system components, including early-life failures,

random failures, and wear-out mechanisms. Its parameters, i.e., the shape parameter (β) and the scale parameter (η), enable the modeling of time-dependent failure rates, which is essential for capturing degradation processes in substation equipment.

Parameter values were identified through a literature review focused on electrical systems and comparable industrial applications, ensuring consistency with the type of equipment and its application within power substations, as well as with reported failure characteristics [28,37]. Table 3 summarizes the Weibull parameters adopted for each component in the RBD model.

Table 3. Weibull parameters for the substation components.

Item #	Component	η (h)	β
1	Grounding Grid	1.47×10^5	1.60
2	Transmission Line	1.36×10^6	1.60
3	Lightning Rods	3.13×10^7	1.60
4	Potential Transformer (PT)	1.72×10^6	2.40
5	Current Transformer (CT)	3.85×10^6	2.40
6	Disconnect Switch	2.19×10^5	1.60
7	Circuit Breakers	3.33×10^6	2.20
8	Standby Set 1	1.01×10^5	2.40
9	Busbars	3.85×10^6	1.60
10	Standby Set 2	1.14×10^5	2.40
11	Single-Phase Power Transformer	4.31×10^5	2.40
12	Three-phase Power Transformer	4.46×10^5	2.40
13	Standby Set 3	1.00×10^9	1.60
14	Auxiliary Power Transformer	2.08×10^6	2.40
15	Motor–Generator Set	7.75×10^5	2.23
16	Rectifier	7.69×10^5	1.42
17	Battery Bank	1.00×10^5	1.70
18	Billing Meter 230 kV	1.00×10^9	1.30
19	Billing Meter 138 kV	1.00×10^9	2.30
20	Intelligent Electronic Devices (IED)	2.04×10^5	1.30
21	Human–Machine Interface (HMI)	3.85×10^5	1.30
22	Input and Output (I/O) Module	1.00×10^6	1.30
23	Communication Protocols	1.00×10^9	1.30

For components with negligible influence on system failure behavior or lacking representative data, high characteristic life values ($\eta = 10^9$ h) were assigned. This assumption avoids artificial bias in system reliability estimation while preserving the structural representation of the system.

The adopted parameterization provides a consistent basis for reliability modeling and enables the evaluation of different degradation scenarios. In particular, the use of Weibull parameters allows the assessment of how variations in failure rate behavior and degradation patterns affect system performance and maintenance decision outcomes, as explored in the sensitivity analysis presented in Section 4.

3.5. Maintenance Parameters: Restoration, Repair Time, and Frequency

In this step, the parameters associated with PM and CM were defined based on maintenance procedures, operational records, and engineering assessments, as described in Section 3.3.

PM frequencies were obtained directly from the substation maintenance plan and technical procedures, which define the execution intervals and scope of maintenance

activities for each component. These values provide the basis for representing maintenance policies in the RBD model.

The restoration factor associated with PM represents the effectiveness of maintenance actions in reducing accumulated degradation. This parameter was defined based on the type of intervention and the expected equipment condition after maintenance. It provides a simplified representation of imperfect maintenance, in which components are partially restored rather than returned to an as-good-as-new condition. This modeling approach enables the representation of degradation accumulation over successive maintenance cycles, which is essential for evaluating the impact of maintenance interval extensions.

Repair times were defined based on operational records and maintenance data, complemented by literature values when necessary [34]. These values represent typical intervention durations and were used to parameterize the maintainability model.

Table 4 summarizes the maintenance parameters adopted for each component, including restoration factors, PM frequency, and repair times.

Table 4. Maintenance parameters for the RBD model.

Item #	Component	Estimate Restoration for PM	PM Frequency (Months)	Estimate Repair Time (Hours)
1	Grounding Grid	70%	12	29.4
2	Transmission Line	60%	12	18.0
3	Lightning Rods	70%	72	24.0
4	Potential Transformer (PT)	70%	72	75.9
5	Current Transformer (CT)	70%	72	42.3
6	Disconnect Switch	60%	72	23.0
7	Circuit Breakers	60%	72	9.4
8	Standby Set 1	60%	72	9.4
9	Busbars	90%	72	4.0
10	Standby Set 2	60%	72	2.3
11	Single-Phase Power Transformer	70%	72	75.9
12	Three-phase Power Transformer	70%	72	42.3
13	Standby Set 3	60%	72	9.4
14	Auxiliary Power Transformer	70%	72	6.1
15	Motor–Generator Set	60%	12	48.0
16	Rectifier	100%	4	16.0
17	Battery Bank	100%	4	2.7
18	Billing Meter 230 kV	100%	repair upon failure	48.0
19	Billing Meter 138 kV	100%		48.0
20	Intelligent Electronic Devices (IED)	100%		97.3
21	Human–Machine Interface (HMI)	100%		148.1
22	Input and Output (I/O) Module	100%		48.0
23	Communication Protocols	100%		48.0

The last six components listed in Table 4 correspond to electronic devices for which no periodic PM is defined. Their failure behavior was modeled using the Weibull distribution, and corrective actions were assumed to restore functionality after failure, resulting in negligible impact on system availability within the analyzed time horizons.

3.6. Constructing the RBD

The RBD was constructed based on the functional decomposition of the substation described in Section 3.1, ensuring consistency with the modeling assumptions defined in Section 2.

Each subsystem was represented according to its functional role and contribution to power transmission, with components arranged in series or parallel configurations depending on their operational dependencies and redundancy. This structure allows the

representation of both critical paths and alternative configurations within the system, reflecting the ability of the substation to maintain operation under component failures.

Figure 6 presents a macro view of the RBD, providing an overview of the main system structure without detailing individual component connections. The complete RBD, developed based on detailed system documentation, is presented in Appendix A.

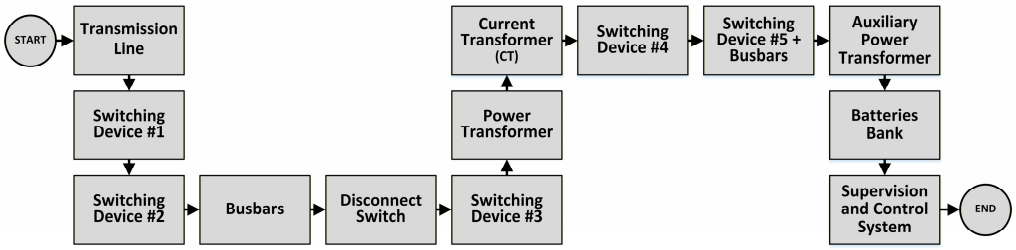


Figure 6. High-level reliability block diagram of the substation.

This representation forms the basis for the evaluation of system reliability, failure occurrence, and the impact of maintenance strategies, as explored in the subsequent analysis.

3.7. Simulation Planning

The simulation study was designed to evaluate the impact of PM interval extensions on system reliability, availability, and failure occurrence using the RBD model. Initially, a baseline maintenance strategy was implemented by incorporating the existing PM schedule for each component into the RBD. This baseline configuration was used to obtain reference values of reliability $R(t)$, availability $A(t)$, and the number of observed failures, which serve as benchmarks for comparison with the proposed extension scenarios.

The analysis considers simulation horizons ranging from 1 to 12 years (8760 to 105,120 h), allowing the evaluation of both short-term and long-term system behavior within the maintenance planning context.

To assess the impact of maintenance interval extension, a set of progressively increasing PM scenarios was defined based on the original maintenance plan. In each scenario, the PM intervals of all components were proportionally extended, as summarized in Table 5. These extensions reflect practical adjustments to the existing maintenance schedule, considering the different original periodicities of the components and operational constraints.

Table 5. PM extension based on the original plan.

Strategy	Grouped Intervals from Original Periodicity (in Months)		
Baseline	4.00	12	72
Extension Plan 1	4.33	13	78
Extension Plan 2	4.67	14	79
Extension Plan 3	5.00	15	80
Extension Plan 4	5.33	16	81
Extension Plan 5	5.67	17	82
Extension Plan 6	6.00	18	83
Extension Plan 7	6.33	19	84

For each scenario, the PM intervals in the RBD were updated, and system performance was evaluated in terms of reliability, availability, and failure occurrence. Results were estimated through repeated stochastic simulations to ensure statistical stability and were compared with the baseline case.

A total of 10,000 Monte Carlo simulation runs were performed for each scenario. A convergence criterion of 1×10^{-6} was adopted, so that the simulations proceeded until

the relative error of the estimated results fell below this threshold. This procedure ensured numerical stability and consistency of the outputs. The random number seed was set to 1, and the simulations were executed on a 16-core computer, allowing parallel processing during the simulation runs.

The evaluation is based on a formal decision criterion defined in Section 2.5, which considers both reliability and failure occurrence relative to the baseline. This approach ensures a consistent and objective assessment of the feasibility of maintenance interval extensions.

4. Results

The initial RBD simulation was performed based on the existing PM plan, which includes scheduled interventions at 4, 12, and 72 months for the respective components (Table 4). The results indicate that, under the current PM strategy, the substation maintains high availability throughout the simulation horizon. As shown in Table 6, the overall substation availability remains consistently above 99.6%, even after 12 years of operation. In contrast, reliability exhibits a continuous decline over time (Table 7), decreasing from approximately 99% in the first year to around 76% after 12 years for the overall system. This divergence highlights how availability and reliability capture fundamentally different aspects of system performance. While availability remains stable due to system redundancy and short repair times, reliability reflects the cumulative effects of degradation and the increasing probability of failure over time.

The subsystem-level analysis indicates that the transmission line is the most critical element in the RBD, presenting the lowest reliability values over time and, consequently, the highest contribution to system failure probability, as shown in Table 7. The Supervision and Control System block appears as the second most critical subsystem, although at a substantially higher reliability level. Therefore, the transmission line represents the dominant contributor to system failure probability and was selected as the focus for evaluating the effects of extending PM intervals. Figure 7 provides an overview of the components that make up the transmission line subsystem in the RBD.

Based on the identification of the transmission line as the most critical subsystem, simulations were conducted to evaluate the feasibility of extending PM intervals. The extension scenarios followed the simulation planning described in Section 3.7 and were defined according to the intervals presented in Table 4. In this stage, the extension was applied to the transmission line to isolate its effect on system behavior.

Table 6. Substation and RBD subsystems block Availability $A(t)$ (%) under the current PM plan.

Simulated Operating Time (Years)	Overall Substation	Transmission Line	Switching Device #1	Switching Device #2	Busbar	Disconnect Switch	Switching Device #3	Power Transformer	Current Transformer (CT)	Switching Device #4	Switching Device #5 + Busbars	Auxiliary Power Transformer	Battery Bank	Supervision and Control System
1	99.99	99.99	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.99	100.00
2	99.81	99.83	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.99	100.00
3	99.76	99.77	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.99	99.99
4	99.74	99.74	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.99	99.99
5	99.72	99.73	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.99	99.99
6	99.71	99.72	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.99	99.99
12	99.63	99.64	99.99	99.99	99.96	99.99	99.99	99.96	99.99	99.99	99.99	99.99	99.99	99.99

Table 7. Substation and RBD subsystems block Reliability $R(t)$ (%) under the current PM plan.

Simulated Operating Time (Years)	Overall Substation	Transmission Line	Switching Device #1	Switching Device #2	Busbar	Disconnect Switch	Switching Device #3	Power Transformer	Current Transformer (CT)	Switching Device #4	Switching Device #5 + Busbars	Auxiliary Power Transformer	Battery Bank	Supervision and Control System
1	98.89	98.89	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
2	97.26	97.26	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
3	94.58	95.54	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.00
4	92.64	93.58	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.00
5	90.42	92.27	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	98.00
6	88.80	90.61	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	98.00
12	76.67	80.72	100.00	100.00	100.00	99.99	99.99	100.00	100.00	100.00	100.00	100.00	100.00	95.00

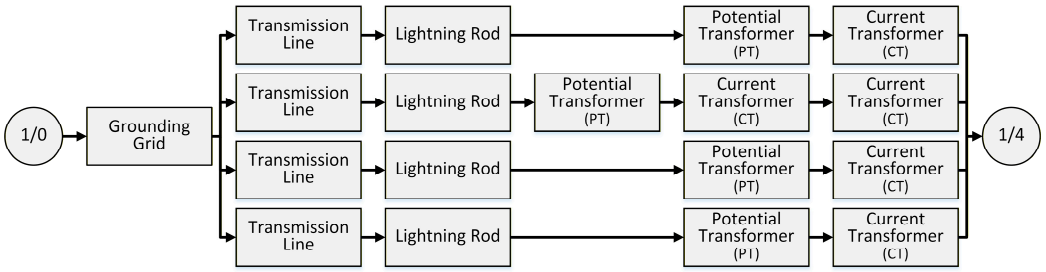


Figure 7. Subsystem transmission line and its components.

The results of the PM extension simulations for the transmission line are summarized in Table 8, which presents the evolution of availability $A(t)$, reliability $R(t)$, and the number of observed failures over the simulation horizon.

Table 8. Simulation results for the transmission line under PM extension scenarios.

Scenario	Metric	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	12 Years
Baseline	$A(t)$	99.99	99.83	99.77	99.74	99.73	99.71	99.64
	$R(t)$	98.89	97.26	95.54	93.58	92.27	90.61	80.72
	Failures	1	3	5	6	8	10	21
Extension Plan 1	$A(t)$	99.99	99.83	99.77	99.74	99.72	99.71	99.64
	$R(t)$	98.89	97.28	95.44	93.69	91.44	90.16	80.86
	Failures	1	3	5	6	9	10	21
Extension Plan 2	$A(t)$	99.99	99.83	99.77	99.74	99.72	99.71	99.64
	$R(t)$	98.89	97.01	95.46	93.72	91.22	89.37	78.62
	Failures	1	3	5	6	9	11	24
Extension Plan 3	$A(t)$	99.99	99.83	99.77	99.74	99.78	99.76	99.66
	$R(t)$	98.89	97.01	95.48	93.73	91.31	88.87	77.77
	Failures	1	3	5	6	9	12	25
Extension Plan 4	$A(t)$	99.96	99.83	99.77	99.82	99.79	99.77	99.68
	$R(t)$	98.89	97.15	95.05	93.25	91.21	88.95	77.75
	Failures	1	3	5	7	9	12	25
Extension Plan 5	$A(t)$	99.99	99.83	99.77	99.82	99.79	99.77	99.69
	$R(t)$	98.89	97.29	95.28	93.33	90.85	88.09	77.33
	Failures	1	3	5	7	9	12	25
Extension Plan 6	$A(t)$	99.99	99.83	99.88	99.82	99.79	99.82	99.71
	$R(t)$	98.89	97.09	95.10	93.12	90.07	88.48	75.56
	Failures	1	3	5	7	10	12	27
Extension Plan 7	$A(t)$	99.99	99.83	99.88	99.83	99.79	99.82	99.71
	$R(t)$	98.89	97.09	95.01	93.31	90.24	88.27	76.20
	Failures	1	3	5	7	10	12	27

A similar trend is observed in the number of failures, which increases with the extension of maintenance intervals, as shown in Table 9. Although the differences between scenarios are relatively small in the early years, they become more evident over time, indicating a cumulative effect of reduced maintenance frequency.

Table 9. Failures observed for each PM extension during simulations.

Scenario	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	12 Years
Baseline	1	3	5	6	8	10	21
Extension Plan 1	1	3	5	6	9	10	21
Extension Plan 2	1	3	5	6	9	11	24
Extension Plan 3	1	3	5	6	9	12	25
Extension Plan 4	1	3	5	7	9	12	25
Extension Plan 5	1	3	5	7	9	12	25
Extension Plan 6	1	3	5	7	10	12	27
Extension Plan 7	1	3	5	7	10	12	27

On the other hand, availability $A(t)$ shows a marginal variation across all scenarios, with a slight increasing trend as maintenance intervals are extended. This behavior reflects the reduction in planned maintenance downtime, which accumulates over time as fewer interventions are performed. Despite this effect, the overall variation in availability remains small throughout the simulation period, indicating that $A(t)$ is relatively insensitive to the changes in maintenance intervals under the modeled conditions, even in the presence of observable differences in reliability and failure behavior.

The observed behavior can be explained by the relationship between failure occurrence and repair duration in the system. Although extending PM intervals increases the probability of failure and the number of observed failures, the corresponding repair times remain relatively short when compared to the overall operating period. As a result, the additional downtime associated with these failures has a limited impact on availability.

At the same time, the reduction in the number of planned maintenance interventions contributes to a slight increase in availability, as fewer scheduled outages are introduced into the system. This combined effect explains why availability remains nearly constant (or even slightly increases) despite the progressive degradation observed in reliability and the increase in failure occurrence.

These results have direct implications for maintenance decision-making. The observed insensitivity of availability to changes in PM intervals indicates that $A(t)$ alone is not sufficient to discriminate between different maintenance strategies in this context. In contrast, reliability $R(t)$ and the number of observed failures provide a more representative measure of the impact of maintenance interval extension on system behavior.

Therefore, the evaluation of the proposed maintenance scenarios must be based on a combined analysis of reliability and failure occurrence, as defined in the decision-making framework presented in Section 2.5. This approach ensures that potential operational benefits associated with extended maintenance intervals are assessed without compromising system performance and risk levels.

To account for the practical relevance of reliability variations, a tolerance parameter ε is introduced in the comparison with the baseline scenario, as defined in Equation (2). This parameter establishes the minimum acceptable reliability level for each simulation horizon, allowing small deviations from the baseline reliability to be considered acceptable without compromising practical decision relevance.

Table 10 presents the minimum acceptable reliability values $R_{lim}(t)$ for different values of ε over the simulation horizon.

Table 10. Minimum acceptable reliability values $R_{lim}(t)$ for different tolerance margins ε.

Tolerance ε	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	12 Years
0	98.89	97.26	95.54	93.58	92.27	90.61	80.72
0.25	98.64	97.01	95.29	93.33	92.02	90.36	80.47
0.5	98.39	96.76	95.04	93.08	91.77	90.11	80.22
1	97.89	96.26	94.54	92.58	91.27	89.61	79.72
2	96.89	95.26	93.54	91.58	90.27	88.61	78.72

The application of the decision-making criteria defined in Section 2.5 to the simulation results presented in Table 8, in conjunction with the reliability thresholds defined in Table 10, enables the identification of maintenance strategies suitable for implementation. Since the simulation horizons represent cumulative system behavior, the evaluation must be interpreted progressively. A maintenance strategy that fails to satisfy the decision criteria at a given horizon is deemed unacceptable for all longer horizons.

The evaluation of the number of observed failures provides a first screening of the candidate maintenance strategies. Table 11 summarizes the acceptability of each scenario based solely on the condition $N_s \leq N_{baseline}$.

Table 11. Decision matrix based on the number of failures criterion.

Scenario	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	12 Years
Extension Plan 1	✔	✔	✔	✔	✔	✔	✔
Extension Plan 2	✔	✔	✔	✔	×	—	—
Extension Plan 3	✔	✔	✔	✔	×	—	—
Extension Plan 4	✔	✔	✔	✔	×	—	—
Extension Plan 5	✔	✔	✔	×	—	—	—
Extension Plan 6	✔	✔	✔	×	—	—	—
Extension Plan 7	✔	✔	✔	×	—	—	—

✔ Acceptable; × Not acceptable; — Not evaluated (cumulative rejection).

The results show that this criterion establishes a structural limit for each strategy. Plans 2 to 4 remain acceptable up to 4 years, while Plans 5 to 7 are limited to 3 years. Plan 1 is the only strategy that remains acceptable over the entire simulation horizon.

The decision matrix for $\varepsilon = 1$ is presented in Table 12. Under this condition, all strategies remain acceptable up to 4 years. Beyond this point, the strategies begin to diverge: Extension Plan 3 remains acceptable up to 5 years, while Extension Plans 2 and 4 to 7 are rejected after 4 years. Extension Plan 1 exhibits the most robust behavior, remaining acceptable over the full 12-year simulation horizon.

Table 12. Decision matrix based on the reliability criterion for $\varepsilon = 1$.

Scenario	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	12 Years
Extension Plan 1	✔	✔	✔	✔	✔	✔	✔
Extension Plan 2	✔	✔	✔	✔	×	—	—
Extension Plan 3	✔	✔	✔	✔	✔	×	—
Extension Plan 4	✔	✔	✔	✔	×	—	—
Extension Plan 5	✔	✔	✔	✔	×	—	—
Extension Plan 6	✔	✔	✔	✔	×	—	—
Extension Plan 7	✔	✔	✔	✔	×	—	—

✔ Acceptable; × Not acceptable; — Not evaluated (cumulative rejection).

The influence of the tolerance parameter ε on the decision outcomes is summarized in Table 13, which presents the maximum acceptable simulation horizon for each maintenance strategy under different tolerance levels. This consolidated representation enables the combined interpretation of the reliability and failure criteria across the entire range of ε values.

Table 13. Maximum acceptable simulation horizon under the reliability criterion for different tolerance levels.

Scenario	$\varepsilon = 0$	$\varepsilon = 0.25$	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$
Extension Plan 1	2 years	4 years	4 years	12 years	12 years
Extension Plan 2	1 year	4 years	4 years	4 years	6 years
Extension Plan 3	1 year	4 years	4 years	5 years	6 years
Extension Plan 4	1 year	2 years	4 years	4 years	6 years
Extension Plan 5	1 year	2 years	4 years	4 years	6 years
Extension Plan 6	1 year	2 years	4 years	4 years	4 years
Extension Plan 7	1 year	2 years	2 years	4 years	4 years

The results show that increasing ε expands the feasible operating region, allowing maintenance strategies with small deviations in reliability to remain acceptable over longer horizons. However, the relative ranking of the strategies remains consistent across all values of ε.

Extension Plan 1 consistently exhibits the most robust performance, remaining acceptable over the longest horizons regardless of the tolerance level. Plans 2 to 4 show intermediate behavior, while plans 5 to 7 are limited to shorter acceptable horizons, even for higher values of ε.

These results indicate that ε primarily controls the strictness of the reliability constraint, without altering the overall ranking of the maintenance strategies, demonstrating the robustness of the proposed decision-making framework.

The combined effect of the reliability and failure criteria is summarized in Table 14, which presents the maximum acceptable simulation horizon for each maintenance strategy under different tolerance levels. This representation directly reflects the feasible operating region defined by the decision-making framework.

Table 14. Maximum acceptable simulation horizon for each maintenance strategy under combined decision criteria for different tolerance levels.

Scenario	$\varepsilon = 0$	$\varepsilon = 0.25$	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$
Extension Plan 1	2 years	4 years	4 years	12 years	12 years
Extension Plan 2	1 year	4 years	4 years	4 years	4 years
Extension Plan 3	1 year	4 years	4 years	4 years	4 years
Extension Plan 4	1 year	2 years	4 years	4 years	4 years
Extension Plan 5	1 year	2 years	3 years	3 years	3 years
Extension Plan 6	1 year	2 years	3 years	3 years	3 years
Extension Plan 7	1 year	2 years	2 years	2 years	2 years

The results show that the acceptable horizon is primarily constrained by the failure criterion for more aggressive maintenance strategies, while the tolerance parameter ε mainly affects the acceptable range for moderate extensions. Extension Plan 1 consistently provides the largest feasible horizon, while plans 5 to 7 exhibit significantly more restrictive applicability.

The final decision results can be interpreted considering the original maintenance periodicities of the transmission line components and the structure of the proposed exten-

sion plans. As shown in Table 3, the relevant components are subjected to PM cycles of 12 and 72 months. The extension strategies defined in Table 4 systematically shift these periodicities, increasing the annual cycle from 12 to 13–19 months and the six-year cycle from 72 to 78–84 months.

These extensions act by delaying discrete maintenance events rather than modifying a continuous process. In short and intermediate horizons, system behavior is primarily governed by the postponement of annual maintenance actions. Since these interventions occur more frequently, even small extensions accumulate rapidly, directly affecting reliability and failure occurrence. This explains why most strategies remain acceptable over intermediate horizons but begin to diverge as delays accumulate.

In longer horizons, the effect of extending the 72-month interventions becomes dominant. As the simulation period approaches and exceeds this threshold, major maintenance actions are significantly delayed, leading to a sharper degradation in reliability and a corresponding increase in failures.

This combined effect explains the progressive loss of acceptability for higher extension levels. Moderate extensions remain viable while system behavior is dominated by the annual cycle, whereas more aggressive strategies, affecting both maintenance cycles, exhibit a more limited applicability.

From a practical perspective, PM extensions must be evaluated with respect to the underlying maintenance periodicities. Small extensions can be accommodated under shorter cycles, while larger extensions become critical when they delay infrequent but high-impact interventions, such as those scheduled every 72 months.

Based on these results, Extension Plan 1 emerges as the only strategy that remains consistently acceptable across all tolerance levels and simulation horizons for this case study. It satisfies both the reliability and failure-based decision criteria in a stable manner, whereas all other strategies exhibit restricted applicability.

While the previous analysis addressed the influence of the tolerance parameter ε, it is also necessary to evaluate the sensitivity of the results to uncertainty in the Weibull parameters used to model component degradation. Therefore, a sensitivity analysis is performed to assess the robustness of the predicted system behavior and decision outcomes.

4.1. Sensitivity Analysis

The sensitivity analysis results are presented considering simultaneous $\pm 20\%$ variations applied to the Weibull parameters relative to their nominal values in Table 2. In this analysis, both the scale parameter η and the shape parameter β are varied proportionally and in the same direction.

Table 15 summarizes these variations for the transmission line components, ensuring a consistent modification of both the characteristic life and the degradation profile across all elements.

To interpret these variations, it is important to note that the scale parameter η governs the characteristic life of the components, while the shape parameter β controls the rate of degradation. As both parameters are varied simultaneously in the same direction, their effects partially counteract each other: changes in η shift the failure distribution in time, whereas changes in β affect the rate of failure acceleration.

Three scenarios were evaluated: (i) baseline, with original parameter values; (ii) lower-bound, with η and β reduced by 20%; and (iii) upper-bound, with η and β increased by 20%. For each scenario, the simulation procedure described in Section 2.4 was applied, and system performance was evaluated in terms of reliability $R(t)$, availability $A(t)$, and the number of observed failures.

Table 15. Weibull parameters for transmission line components and corresponding ±20% variations applied in the sensitivity analysis.

Item #	Component	η (h) [−20%]	η (h) [Nominal]	η (h) [+20%]	β [−20%]	β [Nominal]	β [+20%]
1	Grounding Grid	1.18×10^5	1.47×10^5	1.76×10^5	1.28	1.60	1.92
2	Transmission Line	1.09×10^6	1.36×10^6	1.63×10^6	1.28	1.60	1.92
3	Lightning Rods	2.50×10^7	3.13×10^7	3.75×10^7	1.28	1.60	1.92
4	Potential Transformer (PT)	1.38×10^6	1.72×10^6	2.07×10^6	1.92	2.40	2.88
5	Current Transformer (CT)	3.08×10^6	3.85×10^6	4.62×10^6	1.92	2.40	2.88

The simulation results obtained for the sensitivity analysis are presented in Tables 16 and 17, corresponding to the lower-bound (−20%) and upper-bound (+20%) parameter variations, respectively.

A clear asymmetry is observed in the system response. In the lower-bound scenario (Table 16), a significant reduction in reliability is observed across all maintenance strategies, particularly over longer horizons. For instance, the baseline scenario exhibits a sharp decrease in $R(t)$, reaching values below 60% after 12 years, accompanied by a substantial increase in the number of observed failures. In contrast, the upper-bound scenario (Table 17) shows a consistent improvement in reliability levels and a reduction in failure occurrence across all strategies.

Table 16. Simulation results for the transmission line under Weibull parameter variation (−20%).

Scenario	Metric	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	12 Years
Baseline	$A(t)$	100.00	99.80	99.80	99.70	99.70	99.70	99.60
	$R(t)$	96.70	92.50	88.10	84.30	80.20	76.80	57.90
	Failures	3	8	13	17	22	26	54
Extension Plan 1	$A(t)$	100.00	99.80	99.80	99.70	97.70	99.70	99.60
	$R(t)$	96.70	92.40	87.90	84.60	80.00	77.10	57.20
	Failures	3	8	13	17	22	25	54
Extension Plan 2	$A(t)$	100.00	99.80	99.80	99.70	99.70	99.70	99.60
	$R(t)$	96.70	92.40	87.90	84.30	79.70	76.50	57.50
	Failures	3	8	13	17	22	26	54
Extension Plan 3	$A(t)$	100.00	99.80	99.80	99.70	99.80	99.80	99.70
	$R(t)$	96.70	92.50	87.70	84.20	79.60	75.90	56.90
	Failures	3	8	13	17	22	27	56
Extension Plan 4	$A(t)$	100.00	99.80	99.80	99.80	99.80	99.80	99.70
	$R(t)$	96.70	92.60	87.30	83.30	79.60	75.60	55.30
	Failures	3	8	13	18	23	28	58
Extension Plan 5	$A(t)$	100.00	99.80	99.80	99.80	99.80	99.80	99.70
	$R(t)$	96.70	92.70	87.10	82.80	79.50	75.50	54.90
	Failures	3	8	13	19	22	28	59
Extension Plan 6	$A(t)$	100.00	99.80	99.90	99.80	99.80	99.80	99.70
	$R(t)$	96.70	92.70	87.10	82.70	79.50	75.20	53.40
	Failures	3	8	14	19	23	28	60
Extension Plan 7	$A(t)$	100.00	99.80	99.90	99.80	99.80	99.80	99.70
	$R(t)$	96.70	92.60	86.80	82.60	78.60	75.40	54.40
	Failures	3	8	14	19	24	28	59

Table 17. Simulation results for the transmission line under Weibull parameter variation (+20%).

Scenario	Metric	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	12 Years
Baseline	$A(t)$	100.00	99.80	99.80	99.70	99.70	99.70	99.60
	$R(t)$	100.00	99.20	98.70	97.80	97.10	96.60	92.70
	Failures	0	1	1	2	3	3	8
Extension Plan 1	$A(t)$	100.00	99.80	99.80	99.80	99.70	99.70	99.60
	$R(t)$	100.00	99.10	98.60	97.90	97.00	96.30	92.00
	Failures	0	1	1	2	3	4	8
Extension Plan 2	$A(t)$	100.00	99.80	99.80	99.80	99.70	99.70	99.60
	$R(t)$	100.00	99.10	98.40	97.90	97.00	96.30	91.70
	Failures	0	1	2	2	3	4	9
Extension Plan 3	$A(t)$	100.00	99.80	99.80	99.80	99.80	99.80	99.70
	$R(t)$	100.00	99.00	98.40	97.70	96.90	95.60	90.80
	Failures	0	1	2	2	3	4	9
Extension Plan 4	$A(t)$	100.00	99.80	99.80	99.80	99.80	99.80	99.70
	$R(t)$	100.00	99.00	98.40	97.40	96.60	95.40	91.50
	Failures	0	1	2	3	3	5	10
Extension Plan 5	$A(t)$	100.00	99.80	99.80	99.80	99.80	99.80	99.70
	$R(t)$	100.00	99.00	98.40	97.20	96.50	95.50	90.20
	Failures	0	1	2	3	4	5	10
Extension Plan 6	$A(t)$	100.00	99.80	99.90	99.80	99.80	99.80	99.70
	$R(t)$	100.00	99.10	98.10	97.30	96.50	95.60	89.70
	Failures	0	1	2	3	4	4	11
Extension Plan 7	$A(t)$	100.00	99.80	99.90	99.80	99.80	99.80	99.70
	$R(t)$	100.00	99.10	99.10	97.30	96.30	95.30	89.60
	Failures	0	1	2	3	4	5	11

However, despite these changes in absolute values, the relative differences between the maintenance strategies remain largely unchanged. The gap between scenarios is preserved across both parameter variations, indicating that the overall behavior of the strategies is shifted but not fundamentally altered.

The decision matrices obtained under Weibull parameter variations are presented in Tables 18–21, considering both the number of failures and reliability criteria for the lower-bound (−20%) and upper-bound (+20%) scenarios.

Table 18. Decision matrix based on the number of failures criterion (−20%).

Scenario	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	12 Years
Extension Plan 1	✔	✔	✔	✔	✔	✔	✔
Extension Plan 2	✔	✔	✔	✔	✔	✔	✔
Extension Plan 3	✔	✔	✔	✔	✔	✔	✔
Extension Plan 4	✔	✔	✔	✔	✔	×	—
Extension Plan 5	✔	✔	✔	×	—	—	—
Extension Plan 6	✔	✔	✔	×	—	—	—
Extension Plan 7	✔	✔	×	—	—	—	—

✔ Acceptable; × Not acceptable; — Not evaluated (cumulative rejection).

In the lower-bound scenario (Tables 18 and 20), a strong alignment is observed between the two decision criteria. Both reliability and the number of failures lead to consistent acceptable horizons across all maintenance strategies. Extension Plans 1 to 3 remain acceptable over the full simulation horizon, while more aggressive strategies are progressively

restricted. This indicates that, under more severe degradation conditions, both criteria provide a coherent basis for decision-making.

Table 19. Decision matrix based on the number of failures criterion (+20%).

Scenario	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	12 Years
Extension Plan 1	✔	✔	✔	✔	✔	✔	✔
Extension Plan 2	✔	✔	✔	✔	✔	×	—
Extension Plan 3	✔	✔	×	—	—	—	—
Extension Plan 4	✔	✔	×	—	—	—	—
Extension Plan 5	✔	✔	×	—	—	—	—
Extension Plan 6	✔	✔	×	—	—	—	—
Extension Plan 7	✔	✔	×	—	—	—	—

✔ Acceptable; × Not acceptable; — Not evaluated (cumulative rejection).

Table 20. Decision matrix based on the reliability criterion for $\varepsilon = 1$ (−20%).

Scenario	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	12 Years
Extension Plan 1	✔	✔	✔	✔	✔	✔	✔
Extension Plan 2	✔	✔	✔	✔	✔	✔	✔
Extension Plan 3	✔	✔	✔	✔	✔	✔	✔
Extension Plan 4	✔	✔	✔	✔	✔	×	—
Extension Plan 5	✔	✔	✔	✔	✔	×	—
Extension Plan 6	✔	✔	✔	×	—	—	—
Extension Plan 7	✔	✔	×	—	—	—	—

✔ Acceptable; × Not acceptable; — Not evaluated (cumulative rejection).

Table 21. Decision matrix based on the reliability criterion for $\varepsilon = 1$ (+20%).

Scenario	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	12 Years
Extension Plan 1	✔	✔	✔	✔	✔	✔	✔
Extension Plan 2	✔	✔	✔	✔	✔	✔	✔
Extension Plan 3	✔	✔	✔	✔	✔	✔	×
Extension Plan 4	✔	✔	✔	✔	✔	×	—
Extension Plan 5	✔	✔	✔	✔	✔	×	—
Extension Plan 6	✔	✔	✔	✔	✔	✔	×
Extension Plan 7	✔	✔	✔	✔	✔	×	—

✔ Acceptable; × Not acceptable; — Not evaluated (cumulative rejection).

In contrast, the upper-bound scenario (Tables 19 and 21) reveals a clear divergence between the criteria. While the reliability criterion remains relatively permissive, the number of failures becomes more restrictive, leading to earlier rejection of several strategies. Particularly, plans 3 to 7 are rejected as early as 3 years based on the failure criterion, despite remaining acceptable for longer periods under the reliability criterion.

Compared to the nominal case (Tables 11 and 12), these results highlight a shift in the relative influence of the decision criteria. Under the upper-bound case (+20%), reliability becomes less discriminative, while the number of failures dominates the decision. Conversely, under the lower-bound case (−20%), both criteria converge and lead to consistent outcomes.

Despite this shift, the overall ranking of the maintenance strategies remains unchanged. Extension Plan 1 is consistently identified as the most robust strategy, remaining acceptable over the full simulation horizon in all scenarios, while more aggressive plans exhibit limited applicability.

These results indicate that the proposed decision-making framework is robust with respect to parameter uncertainty, while the relative importance of reliability and failure occurrence depends on the underlying degradation regime.

To consolidate the combined effect of the reliability and failure-based decision criteria under parameter uncertainty, Table 22 summarizes the maximum acceptable simulation horizon for each maintenance strategy considering both criteria simultaneously. The results are presented for the lower-bound (−20%), nominal, and upper-bound (+20%) scenarios of the Weibull parameters.

Table 22. Maximum acceptable simulation horizon for each maintenance strategy under combined decision criteria for different η and β values.

Scenario	−20%	Nominal	+20%
Extension Plan 1	12 years	12 years	12 years
Extension Plan 2	12 years	4 years	5 years
Extension Plan 3	12 years	4 years	2 years
Extension Plan 4	5 years	4 years	2 years
Extension Plan 5	3 years	3 years	2 years
Extension Plan 6	3 years	3 years	2 years
Extension Plan 7	2 years	2 years	2 years

This representation reflects the final decision outcome of the proposed framework, in which a maintenance strategy is considered acceptable only if it satisfies both the reliability and failure criteria. Accordingly, the reported values correspond to the most restrictive acceptable horizon obtained from the intersection of the two criteria for each scenario.

The results in Table 22 provide a consolidated view of the decision-making process, showing that the combined criteria lead to more restrictive outcomes than those obtained from each criterion individually.

A consistent pattern is observed across all scenarios. Extension Plan 1 remains acceptable over the full simulation horizon, confirming its robustness regardless of parameter variation, whereas more aggressive strategies exhibit progressively reduced acceptable horizons.

The influence of Weibull parameter variation becomes evident when comparing the scenarios. In the lower-bound case (−20%), moderately aggressive strategies (Plans 2 and 3) remain acceptable over longer horizons. In contrast, in the upper-bound scenario (+20%), acceptable horizons are significantly reduced, with most strategies beyond Plan 2 limited to short-term horizons, primarily due to the more restrictive failure criterion.

Compared to the nominal case, these results indicate that parameter variations affect the margins of acceptability rather than the overall decision structure. The relative ranking of the strategies is preserved, and Extension Plan 1 consistently exhibits the highest robustness.

Although only proportional variations in η and β were considered, the results provide insight into expected behavior under independent parameter variations. More favorable conditions would be associated with higher η and lower β, whereas the opposite combination would lead to more restrictive outcomes. These cases were not explicitly evaluated and should be addressed in future work.

4.2. Qualitative Cost–Benefit Analysis

From an economic perspective, PM policies influence multiple cost components, including maintenance effort, failure occurrence, system downtime, and operational risk. In practical applications, obtaining reliable monetary estimates for all these components is often difficult, particularly in complex systems with significant indirect effects and

dependencies. Therefore, a structured qualitative cost–benefit analysis is adopted. This analysis is intended as a qualitative interpretation of the simulation results, rather than a detailed economic model.

The total expected cost associated with a maintenance strategy can be conceptually decomposed into its main components, as expressed in Equation (3):

$$C_{total} = C_{PM} + C_{CM} + C_{DT} + C_{risk} \tag{3}$$

where C_{PM}, C_{CM}, C_{DT}, and C_{risk} represent the costs associated with preventive maintenance, corrective maintenance (failures), downtime, and operational risk, respectively. These components can be directly related to the simulation outputs, enabling a consistent qualitative comparison without requiring explicit monetary parametrization.

The simulation results provide a direct interpretation of how maintenance policies affect these cost drivers. Increasing the PM interval reduces the number of preventive actions and, consequently, C_{PM}. However, it increases the number of failures, impacting C_{CM}, and affects availability $A(t)$, which is associated with downtime costs C_{DT}. Reliability $R(t)$, in turn, reflects the probability of failure-free operation and is interpreted as an indicator of operational risk C_{risk}. These relationships define a fundamental trade-off between reduced maintenance effort and increased exposure to failure and risk.

The results show that, for the transmission line subsystem, most extended maintenance strategies are not acceptable under the combined decision criteria defined in Section 2.5. Only Extension Plan 1 remains feasible across all evaluated scenarios, including variations in the Weibull parameters η and β. Therefore, the economic analysis is focused on comparing the baseline strategy with Extension Plan 1.

For a 12-year simulation horizon, the results indicate that failure occurrence remains unchanged between the baseline and Extension Plan 1, availability is preserved, and reliability is slightly improved under Extension Plan 1, while the number of preventive maintenance actions is reduced. These relationships, summarized in Equation (4), indicate that Extension Plan 1 leads to a non-deteriorating or strictly favorable effect across all cost components. Accordingly, the total cost associated with this strategy can be considered lower than or, at worst, equivalent to that of the baseline, without compromising system performance.

$$\begin{aligned} C_{PM}^{(1)} &< C_{PM}^{(0)} \\ C_{CM}^{(1)} &\approx C_{CM}^{(0)} \\ C_{DT}^{(1)} &\approx C_{DT}^{(0)} \\ C_{risk}^{(1)} &\leq C_{risk}^{(0)} \end{aligned} \tag{4}$$

where superscripts (0) and (1) denote the baseline and Extension Plan 1, respectively.

More aggressive maintenance extensions, although reducing C_{PM} and C_{DT}, lead to systematic increases in both C_{CM} and C_{risk}. As a result, these strategies violate the decision criteria and are not viable, regardless of their apparent economic benefits. This demonstrates that reductions in maintenance effort alone are not sufficient to justify PM interval extension, which must satisfy both reliability and failure constraints.

Considering the nominal Weibull parameters and $\varepsilon = 1$ as a representative tolerance level, Extension Plan 1 emerges as the most robust and economically viable candidate. It reduces maintenance effort while maintaining failure occurrence at baseline levels and preserving acceptable reliability. Its acceptability remains stable under all analyzed parameter variations, indicating robustness with respect to uncertainty in the degradation model. Within the proposed framework, Extension Plan 1 represents a consistent trade-off between cost reduction, system performance, and robustness for PM interval extension in the transmission line subsystem.

5. Conclusions

This work proposed a structured decision-making framework for evaluating the extension of PM intervals in complex engineering systems, integrating reliability analysis, failure occurrence, and simulation-based evaluation within a unified approach. The framework combines reliability and failure criteria to identify feasible maintenance strategies under defined acceptance conditions.

The application to a transmission line subsystem showed that PM interval extension must be treated as a constrained decision problem. While increasing the PM interval reduces maintenance effort, it also increases exposure to degradation and failure. The acceptable operating region is therefore defined by the interaction between reliability degradation and failure accumulation, rather than by isolated performance indicators.

The results indicate that only one extension strategy remains consistently feasible across all evaluated conditions. Extension Plan 1 provides a stable balance between maintenance interval increase and system performance, satisfying both reliability and failure-based criteria without introducing additional operational risk. In contrast, more aggressive extensions become rapidly unfeasible, with acceptable horizons strongly dependent on the tolerance level and degradation regime.

The sensitivity analysis confirmed the robustness of this result, showing that the relative performance of the maintenance strategies remains consistent under variations in the Weibull parameters. However, the analysis was limited to proportional variations in η and β, and did not consider independent or combined variations, which should be explored in future work.

It is also noted that the adopted Kijima Type II model does not account for maintenance-induced failures. In this sense, the results can be interpreted as conservative, and the benefits of reducing maintenance interventions may be even more pronounced under alternative imperfect maintenance modeling assumptions [4].

From an economic perspective, the qualitative cost–benefit analysis showed that Extension Plan 1 can reduce maintenance effort and associated downtime while maintaining failure occurrence at baseline levels and preserving acceptable reliability. This demonstrates that economically viable maintenance extensions can be achieved without increasing operational risk when appropriate decision criteria are applied.

Finally, the proposed framework can be adapted to systems with more accurate reliability data, supporting continuous improvement in PM planning as new information becomes available. This is particularly relevant in contexts where predictive maintenance is not feasible or not recommended by Reliability-Centered Maintenance (RCM) analyses.

Overall, the proposed framework provides a systematic and practical approach for maintenance decision-making under uncertainty, enabling the identification of strategies that balance performance, risk, and cost consistently and transparently.

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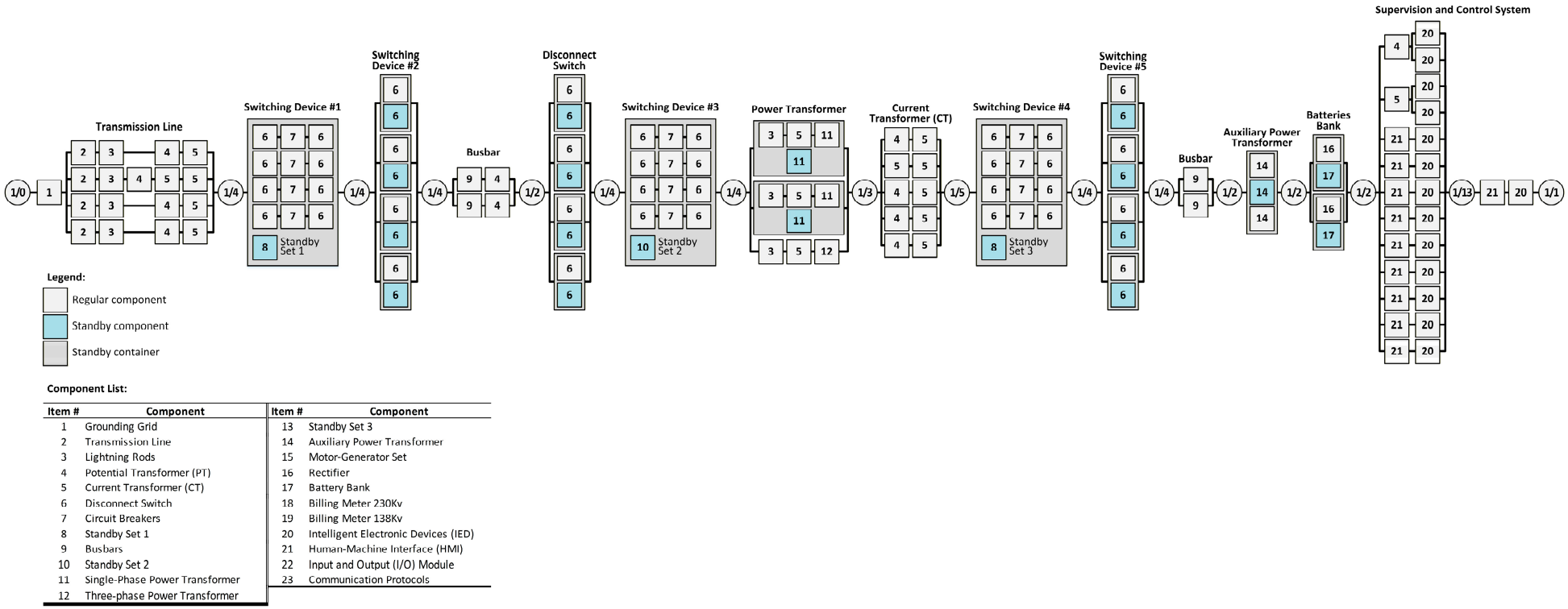
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Abbreviations

The following abbreviations are used in this manuscript:

$A(t)$	Availability
CM	Corrective Maintenance
KPI	Key Performance Indicator
PM	Preventive Maintenance
$R(t)$	Reliability
RBD	Reliability Block Diagram
RCM	Reliability-Centered Maintenance

Appendix A



Component List:

Item #	Component	Item #	Component
1	Grounding Grid	13	Standby Set 3
2	Transmission Line	14	Auxiliary Power Transformer
3	Lightning Rods	15	Motor-Generator Set
4	Potential Transformer (PT)	16	Rectifier
5	Current Transformer (CT)	17	Battery Bank
6	Disconnect Switch	18	Billing Meter 230Kv
7	Circuit Breakers	19	Billing Meter 138Kv
8	Standby Set 1	20	Intelligent Electronic Devices (IED)
9	Busbars	21	Human-Machine Interface (HMI)
10	Standby Set 2	22	Input and Output (I/O) Module
11	Single-Phase Power Transformer	23	Communication Protocols
12	Three-phase Power Transformer		

Figure A1. Substation complete Reliability Block Diagram (RBD).

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